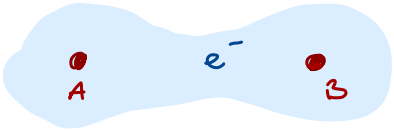
Building molecules

Molecules

simplest case: diatomic molecules

→ H_2^+, HCl, H_2, ...

H_2^+ molecule



$$R = |\underline{r}_B - \underline{r}_A|$$

Schrödinger eqn.

$$\hat{H} = \frac{\hat{p}^2}{2m} + V_A(\underline{r}) + V_B(\underline{r}) + \frac{e^2}{4\pi\epsilon_0}\frac{1}{R}$$

$$-\frac{\hbar^2}{2m}\Delta \qquad \frac{e^2}{4\pi\epsilon_0|\underline{r}-\underline{r}_A|} \qquad \text{proton - proton}$$

$$\hat{H}_e(R)$$

task:

(1) find ground state solution of

$$\hat{H}(R)\cdot\psi(\underline{r}) = E(R)\cdot\psi(\underline{r})$$

depend on R parametrically

(2) find optimal value of R, minimizing E(R)

Variational strategy

statement: ground state wave function ψ of $\hat{H}_e$ minimizes $\langle\hat{H}_e\rangle_\psi$

1. consider trial wave function

$$\psi(\underline{r}) \approx N\left(\varphi_{1s}(\underline{r}-\underline{r}_A) + \varphi_{1s}(\underline{r}-\underline{r}_B)\right)$$

$$= N\cdot(\varphi_A(\underline{r}) + \varphi_B(\underline{r})) \quad \sim \text{real now}$$

2. normalization

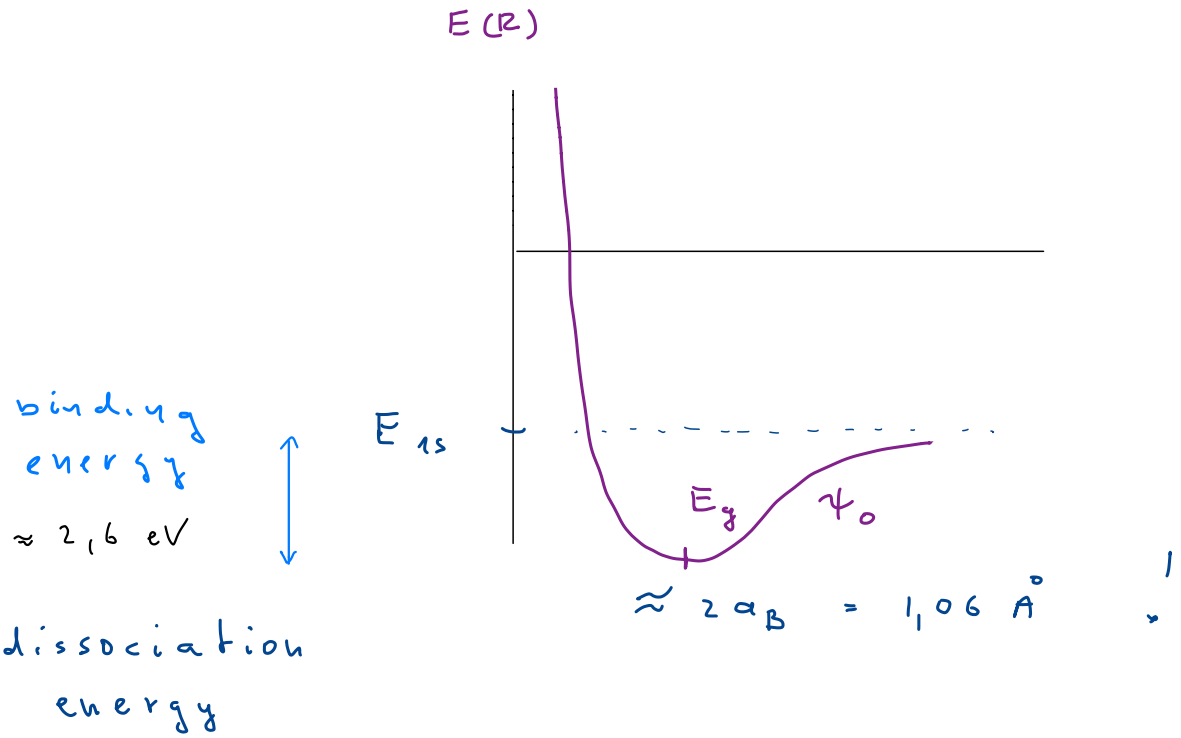
$$\int |\psi|^2(\underline{r})\, d^3\underline{r} = N^2\left(\int_{\underline{r}} \varphi_A^2(\underline{r}) + \int \varphi_B^2(\underline{r}) + 2\int_{\underline{r}} \varphi_A(\underline{r})\,\varphi_B(\underline{r})\right)$$

φ_A^2: $= 1$; φ_B^2: $= 1$; $\varphi_A\varphi_B$: $= S$ (overlap)

$$\Rightarrow \quad N^2 = \frac{1}{2(1+S)} \quad \Rightarrow \quad N = \frac{1}{\sqrt{2}\sqrt{1+S}}$$

3. calculate $\langle \hat{H}_e \rangle_\psi = \ldots = E_e(R)$

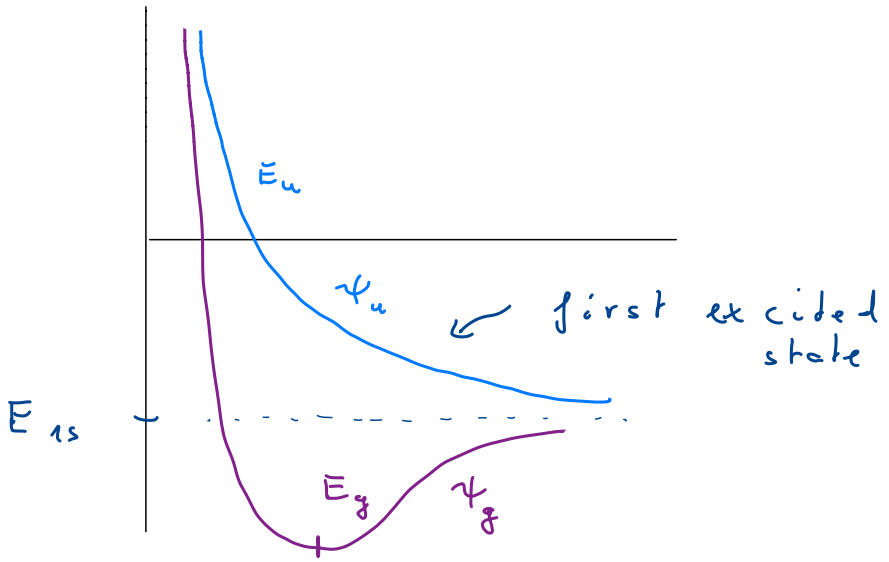
$$E(R) = E_e(R) + \frac{e^2}{4\pi\epsilon_0}\frac{1}{R}$$



$\psi_g \sim \varphi_A + \varphi_B$ ~ symmetrical "gerade"

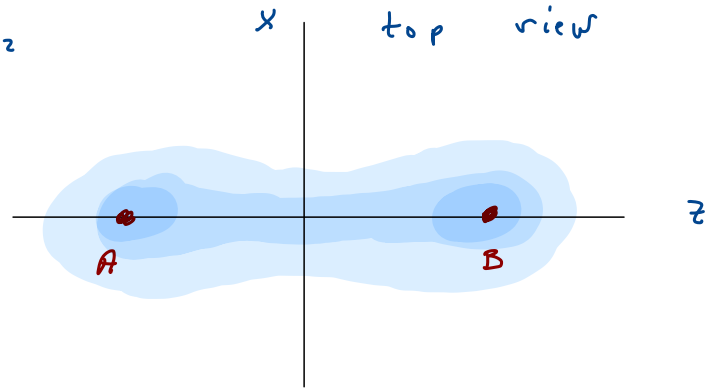
antisymmetrical combination (odd = ungerade)

$\psi_u \sim \varphi_A - \varphi_B$ => larger energy !



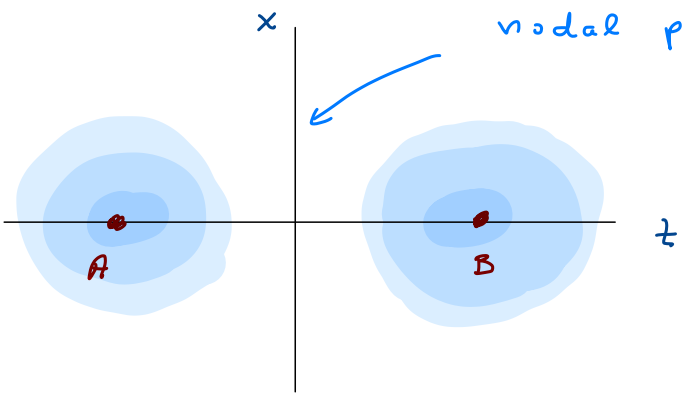
ψ_g ~ bonding molecular orbit

$P_g(r) = |\psi_g(r)|^2$

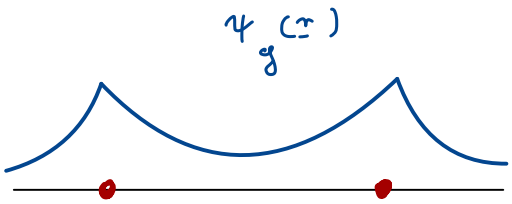


ψ_u ~ antibonding molecular orbit

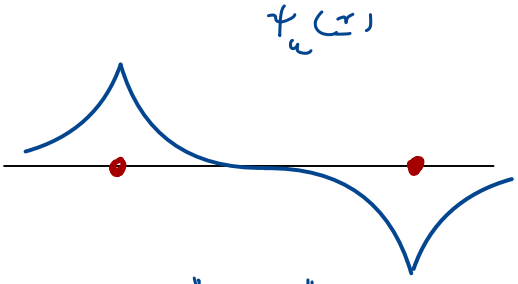
$P_u(r) = |\psi_u(r)|^2$



ψ_g : electrons delocalize → gain kinetic energy ↪ proton - proton repulsion



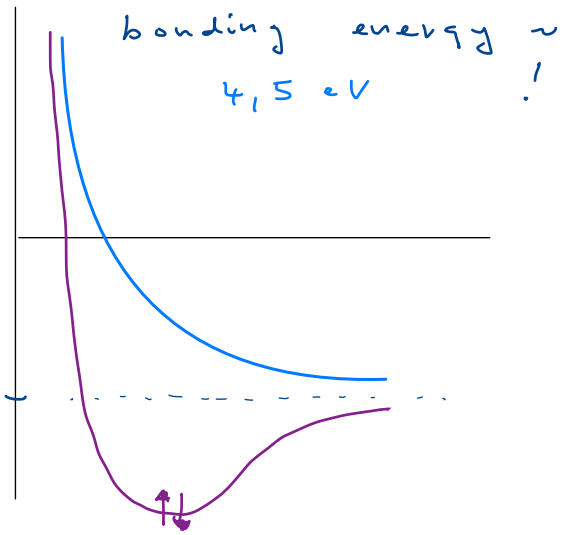
"even"
gerade



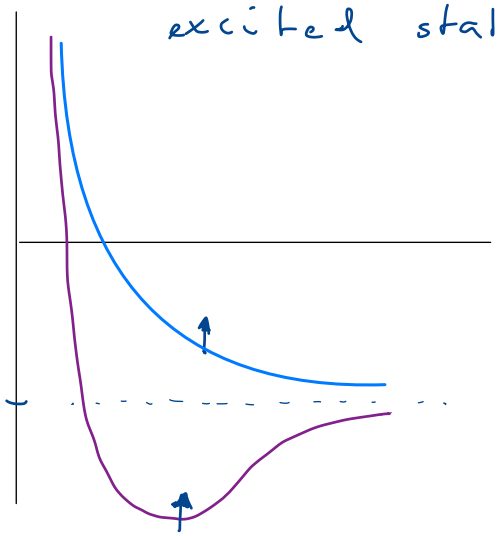
"odd"
ungerade

H_2 molecule =>

$R_{min} \approx 0,74$ Å



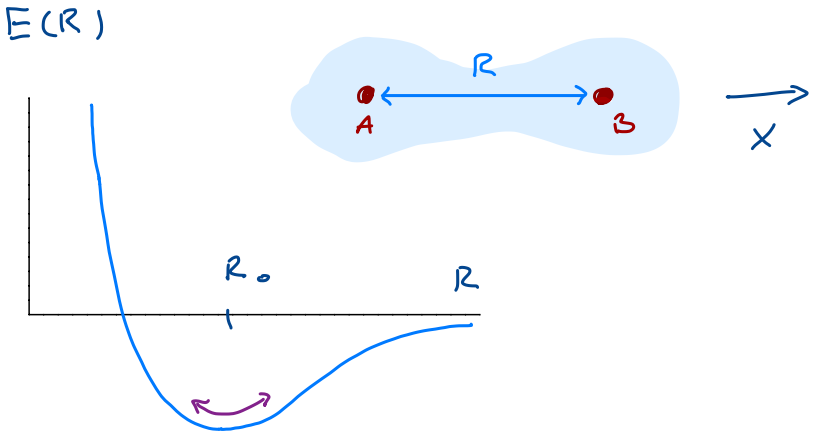
$S = 0$



$S = 1$

($S = 0$)

Vibrational excitations



$$E = E_0 + \frac{1}{2}\underbrace{\frac{\partial^2 E}{\partial R^2}\Big|_{R_0}}_{\equiv D}\cdot (R-R_0)^2 + \dots$$

D ~ effective spring constant

$\Rightarrow$ restoring force $\quad F_A = \frac{\partial E}{\partial R} = D(R-R_0) = -F_B$

$$m_A \ddot{x}_A = D(R-R_0)$$

$$m_B \ddot{x}_B = -D(R-R_0)$$

$$\Rightarrow \quad \ddot{x}_B - \ddot{x}_A = \ddot{R} = -\left(\frac{1}{m_A}+\frac{1}{m_B}\right) D \underbrace{(R-R_0)}_{\equiv u}$$

$$\ddot{u} = \ddot{R}$$

$$\Rightarrow \quad \underbrace{\frac{m_A m_B}{m_A + m_B}}_{m_{eff}} \ddot{u} = -D u$$

m_{eff} effective or reduced mass

$$\boxed{m_{eff}\,\ddot{u} = -D u}$$

$\Rightarrow$ frequency of vibration $\quad \boxed{\omega = \sqrt{\frac{D}{m_{eff}}}}$

simple estimation

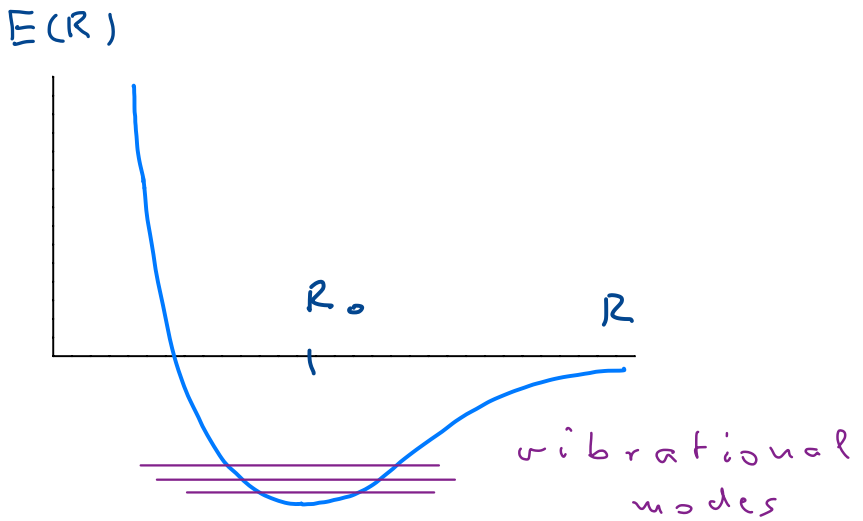
$R_0 \sim 1\,\text{Å}$ $\qquad \frac{\partial^2 E}{\partial R^2} \sim \frac{\partial^2}{\partial R^2}\frac{e^2}{4\pi\epsilon_0 R} = \frac{2e^2}{4\pi\epsilon_0 R_0^3} = D$

$H_2 \rightarrow m_{eff} = \frac{m_p}{2} \approx 1000\, m_e \Rightarrow f \sim 10{,}000\ \text{GHz}$

quantum mechanical oscillations!

$\Rightarrow$ $E_{vib} = \hbar\omega\left(n + \frac{1}{2}\right)$ with $\hbar\omega \sim 0{,}1\ \text{eV}$

for heavier atoms $\sim 0{,}01$ eV



Optical sidebands: Raman scattering

electrons $E_i \rightarrow E_f$

vibrations $n \rightarrow n, n \pm 1$

Rotational excitations

$E_{rot} = \frac{L^2}{2\Theta} = \frac{\hbar^2 \ell(\ell+1)}{2\Theta}$

ℓ ~ rotational Q-number

Θ ~ moment of inertia

$E_{rot} \sim 10^{-4}$ eV

Supplemental info: Two-electron wave functions

One e^- $\Rightarrow$ $\phi_\uparrow(\underline{r})$ and $\phi_\downarrow(\underline{r})$

$|\phi_\uparrow(\underline{r})|^2$ = P(e⁻ at $\underline{r}$ with spin ↑)

$\Rightarrow$ $\phi_\sigma(\underline{r})$ $\sigma = +$ or $\uparrow, \downarrow$

two e^- : $\phi_{\sigma_1\sigma_2}(\underline{r}_1, \underline{r}_2)$ = " $\phi(1,2)$ "

spin of first e⁻ ; its position

$|\phi_{\uparrow\downarrow}(\underline{r}_1, \underline{r}_2)|^2$ = P(first e⁻ at $\underline{r}_1$ and has spin ↑, second at $\underline{r}_2$ and ↓)

2-electron Schrödinger eqn.

$$\hat{H} = \hat{K}_1 + \hat{K}_2 + \hat{V}_{12} =$$

$$= -\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 + \frac{e^2}{4\pi\epsilon_0}\frac{1}{|\underline{r}_1 - \underline{r}_2|}$$

$\Delta_1 = \frac{\partial^2}{\partial \underline{r}_1^2}$ $\quad$ $\Delta_2 = \frac{\partial^2}{\partial \underline{r}_2^2}$

$$\left(-\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 + \frac{e^2}{4\pi\epsilon_0}\frac{1}{|\underline{r}_1 - \underline{r}_2|}\right)\psi_{\sigma_1\sigma_2}(\underline{r}_1, \underline{r}_2) = E \cdot \psi_{\sigma_1\sigma_2}(\underline{r}_1, \underline{r}_2)$$

no analytical solution...

$\Rightarrow$ approximate and numerical solutions

Simplest assumption

Hartree: each e^- has its own state

$$\psi(\underline{r}_1, \underline{r}_2) = \varphi_1(\underline{r}_1) \cdot \varphi_2(\underline{r}_2)$$

examples:

- H_2 ground state: $\sigma_1 = \uparrow \quad \sigma_2 = \downarrow$

$$\phi^g_{\sigma_1\sigma_2}(\underline{r}_1, \underline{r}_2) = \psi_g(\underline{r}_1)\,\psi_g(\underline{r}_2) \cdot \chi_\uparrow(\sigma_1)\,\chi_\downarrow(\sigma_2)$$

- excited state $\sigma_1 = \sigma_2 = \uparrow$

$$\phi^e_{\sigma_1\sigma_2}(\underline{r}_1, \underline{r}_2) = \psi_g(\underline{r}_1) \cdot \psi_u(\underline{r}_2)\,\chi_\uparrow(1)\,\chi_\uparrow(2)$$

Problem:

electrons indistinguishable $\Rightarrow$ $P(1,2) = P(2,1)$

$\phi(1,2) = \phi(2,1)$ bosons

... $\Rightarrow$ $\phi(1,2) = -\phi(2,1)$ fermions (electrons)

wave function is antisymmetrical for fermions

Hartree approximation violates this

$\Rightarrow$ antisymmetrize $\varphi_1(1) \cdot \varphi_2(2)$

$$\psi(1,2) = \frac{1}{\sqrt{2}}\left(\varphi_1(1)\,\varphi_2(2) - \varphi_1(2)\,\varphi_2(1)\right)$$

↑ Slater determinant